

Generative AI - Text Generation and Machine Translation

Assignment (Question & Answer Format)

Question 1: What is Generative AI and what are its primary use cases across industries?

Answer:

Generative AI is a branch of artificial intelligence that learns patterns from existing data and generates new content such as text, images, audio, code, videos, and 3D designs. Unlike traditional AI, which mainly performs prediction or classification, Generative AI creates new outputs that resemble the training data.

Primary use cases:

- Content creation (articles, blogs, marketing)
- Chatbots and virtual assistants
- Code generation
- Healthcare and drug discovery
- Image/video generation
- Education and personalized learning
- Gaming and entertainment
- Product and industrial design.

Question 2: Explain the role of probabilistic modeling in generative models. How do these models differ from discriminative models?

Answer:

Probabilistic modeling enables generative models to learn the probability distribution of data and generate realistic new samples. Generative models learn $P(X)$ or $P(X,Y)$, whereas discriminative models learn $P(Y|X)$. Generative models can create new data, while discriminative models mainly classify or predict labels.

Question 3: What is the difference between Autoencoders and Variational Autoencoders (VAEs) in the context of text generation?

Answer:

Autoencoders learn deterministic latent representations and reconstruct inputs. Variational Autoencoders (VAEs) learn a probability distribution using mean and variance, allowing random sampling from the latent space to generate diverse and meaningful text. VAEs are more suitable for text generation because they can generate multiple valid outputs.

Question 4: Describe the working of attention mechanisms in Neural Machine Translation (NMT). Why are they critical?

Answer:

Attention enables the decoder to focus on the most relevant source words at every decoding step instead of relying on a single context vector. This improves alignment, captures long-range dependencies, increases translation accuracy, and produces better translations for long sentences.

Question 5: What ethical considerations must be addressed when using generative AI for creative content such as poetry or storytelling?

Answer:

Important ethical issues include bias and fairness, copyright and plagiarism, misinformation, deepfakes, privacy, transparency, accountability, harmful content generation, and ensuring human oversight before publication.

Question 6: Use the following small text dataset to train a simple Variational Autoencoder (VAE) for text reconstruction...

Answer:

Explanation: Tokenize the dataset, convert text into sequences, pad sequences, build an encoder-decoder VAE, train it, and reconstruct sentences.

Sample Output:

Input: The sky is blue

Reconstructed: The sky is blue

Generated: The stars are bright

Question 7: Use a pre-trained GPT model to translate a short English paragraph into French and German.

Answer:

Original: Artificial Intelligence is transforming healthcare.

French: L'intelligence artificielle transforme les soins de santé.

German: Künstliche Intelligenz verändert das Gesundheitswesen.

Question 8: Implement a simple attention-based encoder-decoder model for English-to-Spanish translation.

Answer:

Build an LSTM encoder, Bahdanau attention layer, and LSTM decoder with teacher forcing. After training, example output:

English: How are you?

Spanish: ¿Cómo estás?

Question 9: Generate a new poem using a pre-trained GPT model.

Answer:

Prompt: Write a poem similar to the given examples.

Generated Poem:

Golden dawn paints hills with light,
Morning whispers soft and bright.
Dreams awaken with the sun,
Hope begins where day's begun.

Question 10: Design a creative writing assistant using Generative AI.

Answer:

Use GPT-4/Llama/Mistral as the language model with licensed story datasets. Apply instruction tuning and human feedback, perform bias mitigation using filtering and human review, and evaluate using BLEU, ROUGE, perplexity, and human evaluation. Challenges include hallucinations, copyright, privacy, computational cost, and scalability.